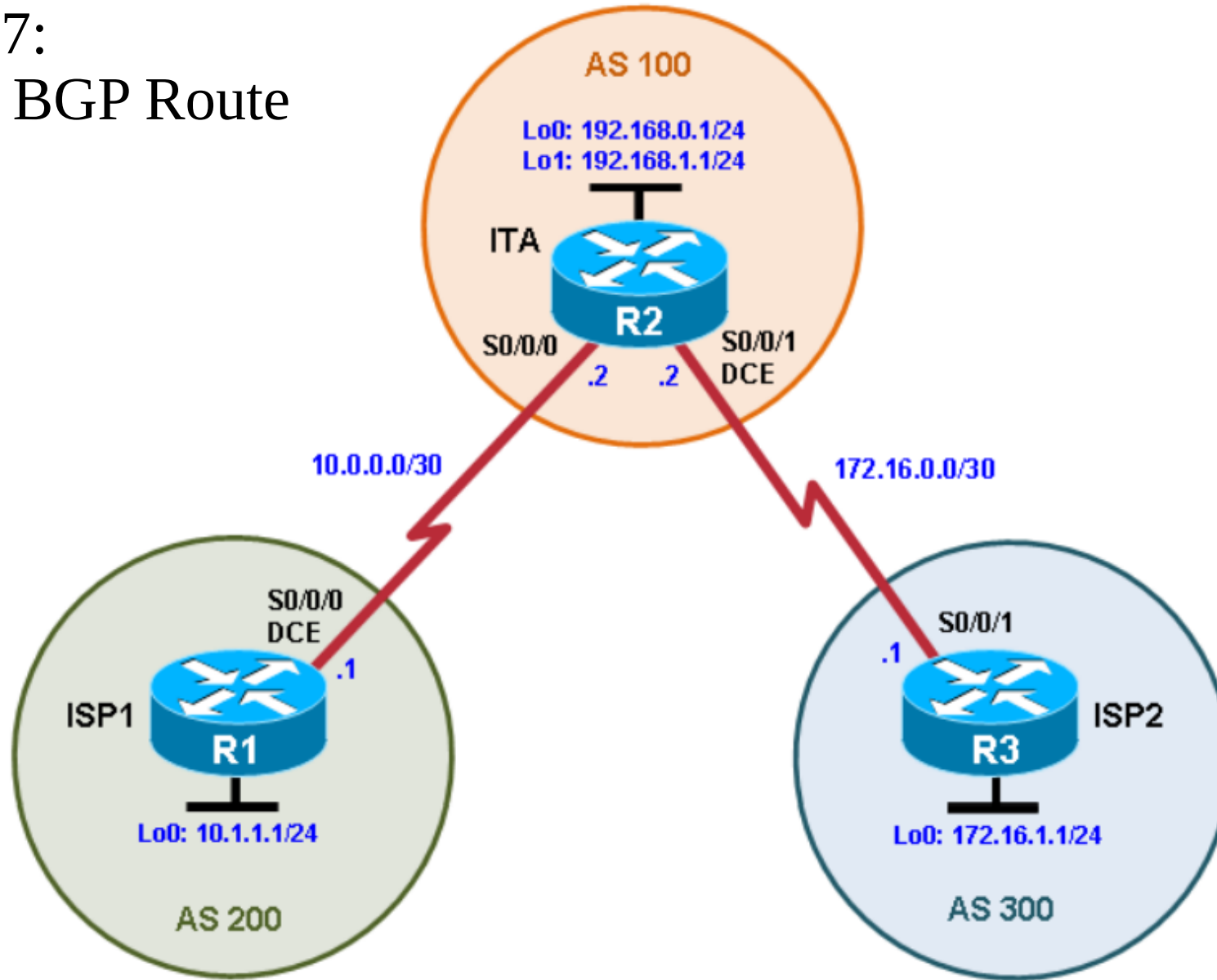


Lab 2.7: Verify BGP Route



Objectives

- Configure BGP to exchange routing information with two ISPs.

Step 1: Configure interface addresses.

```
ISP1(config)# interface Lo0
```

```
ISP1(config-if)# description ISP1 Internet Network
```

```
ISP1(config-if)# ip address 10.1.1.1 255.255.255.0
```

```
ISP1(config-if)# exit
```

```
ISP1(config)# interface Serial0/0/0
```

```
ISP1(config-if)# description ISP1 -> ITA
```

```
ISP1(config-if)# ip address 10.0.0.1 255.255.255.252
```

```
ISP1(config-if)# no shutdown
```

```
ISP1#
```

Step 1: Configure interface addresses.

```
ITA(config)# interface Lo0
```

```
ITA(config-if)# description Core router network link 1
```

```
ITA(config-if)# ip address 192.168.0.1 255.255.255.0
```

```
ITA(config)# exit
```

```
ITA(config-if)# interface Lo1
```

```
ITA(config-if)# description Core router network link 2
```

```
ITA(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
ITA(config-if)# exit
```

Step 1: Configure interface addresses.

```
ITA(config)# interface Serial0/0/0
```

```
ITA(config-if)# description ITA -> ISP1
```

```
ITA(config-if)# ip address 10.0.0.2 255.255.255.252
```

```
ITA(config-if)# no shutdown
```

```
ITA(config-if)# exit
```

```
ITA(config)# interface Serial0/0/1
```

```
ITA(config-if)# description ITA -> ISP2
```

```
ITA(config-if)# ip address 172.16.0.2 255.255.255.252
```

```
ITA(config-if)# no shutdown
```

```
ITA(config-if)# end
```

Step 1: Configure interface addresses.

```
ISP2(config)# interface Lo0
```

```
ISP2(config-if)# description ISP2 Internet Network
```

```
ISP2(config-if)# ip address 172.16.1.1 255.255.255.0
```

```
ISP2(config)# exit
```

```
ISP2(config-if)# interface Serial0/0/1
```

```
ISP2(config-if)# description ISP2 -> ITA
```

```
ISP2(config-if)# ip address 172.16.0.1 255.255.255.252
```

```
ISP2(config-if)# no shutdown
```

```
ISP2(config-if)# end
```

Step 2: Configure BGP on the ISP routers.

- On the ISP1 and ISP2 routers, configure BGP to peer with the ITA boundary router and advertise the ISP loopback networks.

```
ISP1(config)# router bgp 200
```

```
ISP1(config-router)# neighbor 10.0.0.2 remote-as 100
```

```
ISP1(config-router)# network 10.1.1.0 mask 255.255.255.0
```

```
ISP2(config)# router bgp 300
```

```
ISP2(config-router)# neighbor 172.16.0.2 remote-as 100
```

```
ISP2(config-router)# network 172.16.1.0 mask 255.255.255.0
```

Step 3: Configure BGP on the ITA boundary router.

- Configure the ITA router to run BGP with both Internet providers.

```
ITA(config)# router bgp 100
```

```
ITA(config-router)# neighbor 10.0.0.1 remote-as 200
```

```
ITA(config-router)# neighbor 172.16.0.1 remote-as 300
```

```
ITA(config-router)# network 192.168.0.0
```

```
ITA(config-router)# network 192.168.1.0
```


Step 3: Configure BGP on the ITA boundary router.

- To verify the configuration, check the ITA routing table with the show ip route command.

```
ITA# show ip route
```

ITA has BGP routes to the loopback networks at each ISP router